EC2255-LINEAR INTEGRATED CIRCUITS

ASSIGNMENT -MINI PROJECT REPORT

CATEGORY	MARKS ALLOTTED	MARKS AWARDED
Timeliness	4	
Content preparation	4	
Hardware and Software Demo	7	
Team work, Presentation and Viva	5	
TOTAL	20 MARKS	MARKS

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CLASS	:	ECE A
SUBMISSION DATE	:	31.03.2026

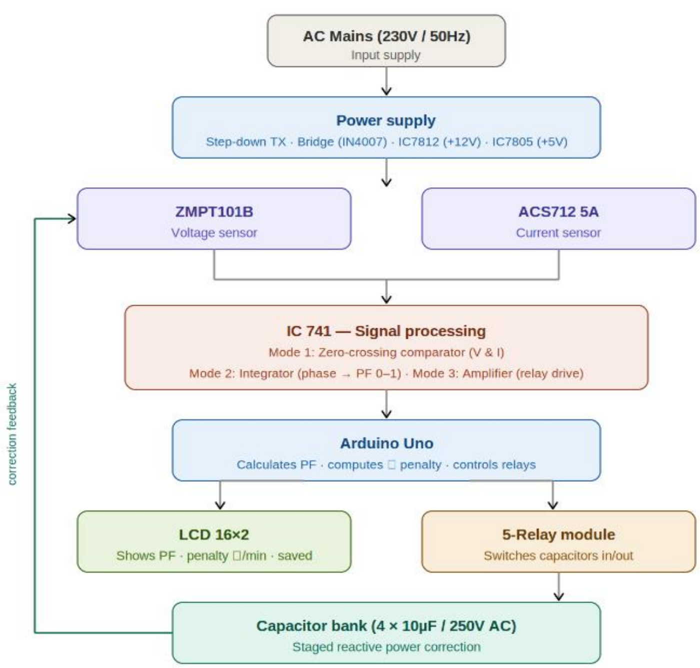
Power Factor Estimation and Correction with Real-Time Energy Tracking

Objectives:

The main objective of the project is to develop an intelligent system that is capable of measuring and improving the power factor of electrical systems. The system utilizes sophisticated sensors such as the ZMPT101B voltage sensor and the ACS712 current sensor to sense AC voltage and current waveforms in real time. The system is also capable of calculating essential electrical parameters such as the power factor, real power, reactive power, and apparent power in real time by processing the signals received from the sensors using microcontrollers such as the Arduino UNO and ESP32.

Another objective of the project is to develop an intelligent system that is capable of automatically improving the power factor of electrical systems. The system is capable of automatically improving the power factor of electrical systems by automatically compensating for the reactive power when the power factor is found to be lower than the set threshold (0.90). The system is also capable of displaying all the real-time parameters such as voltage, current, power factor, and energy consumption wirelessly using an LCD display.

Circuit Diagram:



Theory and Working:

Power Factor and Problem :

The power factor is an important parameter in electrical systems, which measures the efficient utilization of electrical power. It is defined as the ratio of active power in watts to the apparent power in volt-amperes, which can be mathematically expressed as follows:

PF = cos φ, where φ represents the phase angle between voltage and current. For example, in an electrical system, due to the inductive reactance of the motors, the current drawn from the source lags behind the voltage, thus resulting in a low power factor.

The low power factor causes an increase in the amount of current drawn from the source, which in turn leads to transmission losses, poor voltage regulation, and higher electricity bills.

Sensing Voltage and Current:

For safe measurement of the high AC voltage levels, the ZMPT101B voltage sensor module is used. The module steps down the 230V AC supply to a safer level that can be measured and input into the microcontroller. The ACS712 current sensor module is used to measure the current flowing through the load without any direct electrical contact. This ensures safety and accuracy in the measurement process.

The analog signal obtained from the sensor modules is very low and needs conditioning. For this purpose, the IC741 operational amplifier is used. The conditioned signal is then input into the ADC of the Arduino.

Real-Time Computation:

The Arduino UNO constantly measures the voltage and current values. It then uses this information to compute the RMS voltage (VRMS) and the RMS current (IRMS). The Arduino also calculates the phase difference, which is then used to compute the power factor.

The ESP32 microcontroller then carries out more complex operations such as computing active power (P = VRMS * IRMS * PF), reactive power (Q), apparent power (S), and total energy consumption in units of kWh. The computations are carried out in real time, allowing for the constant monitoring of performance.

Automatic Correction:

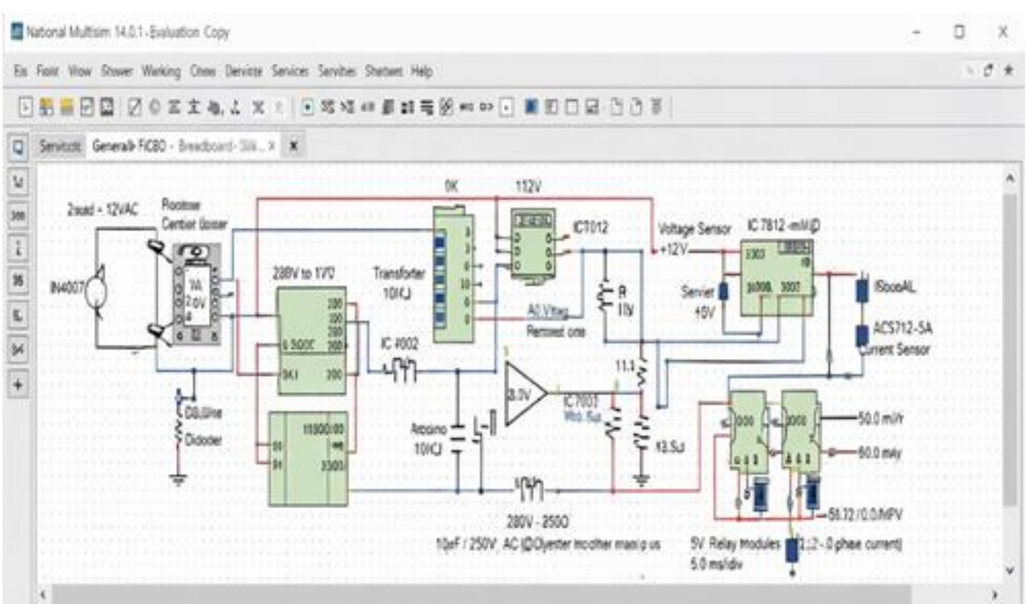
Once the power factor is found to be less than 0.90, corrective action is taken. In this case, relay modules are switched on by the ESP32, which connects capacitors in parallel with the load. These capacitors are used to provide leading power factor, which is used to neutralize the lagging power factor of inductive loads.

This reduces the phase difference between current and voltage, resulting in an improvement in power factor close to unity. The capacitors are used in parallel to provide precise power factor correction.

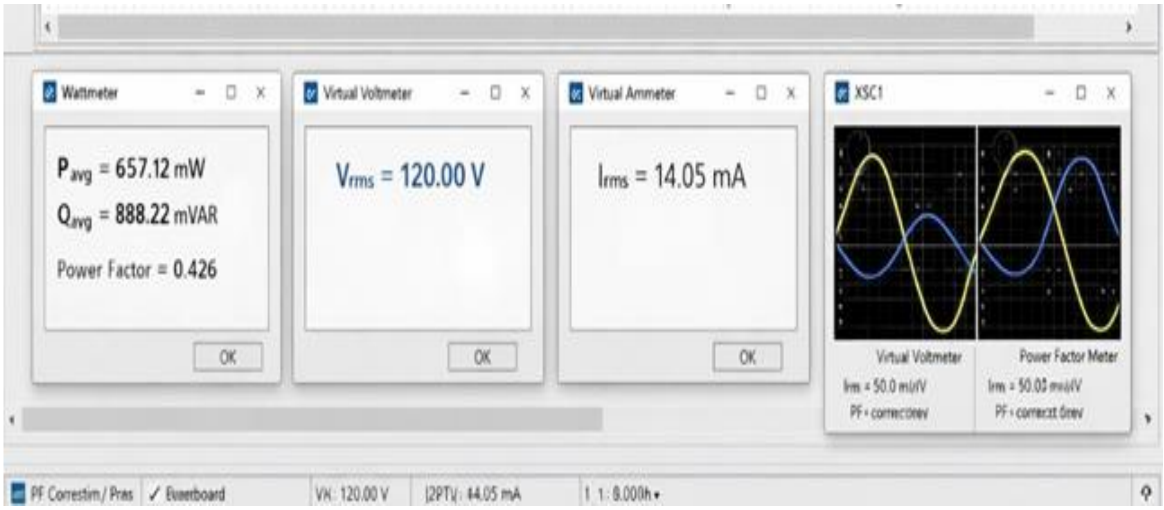
Power Supply:

The system is powered by a regulated DC supply derived from a 230V AC supply. The AC supply is reduced to 12V AC using a transformer and is further converted to a DC supply using a bridge rectifier. The output is filtered using a capacitor and regulated to 12V and 5V using voltage regulators IC 7812 and IC 7805, respectively. These regulated voltages are used to power the sensors, microcontrollers, and other components in the circuit.

<u>Simulated Circuit Diagram:</u>



Simulated Output:



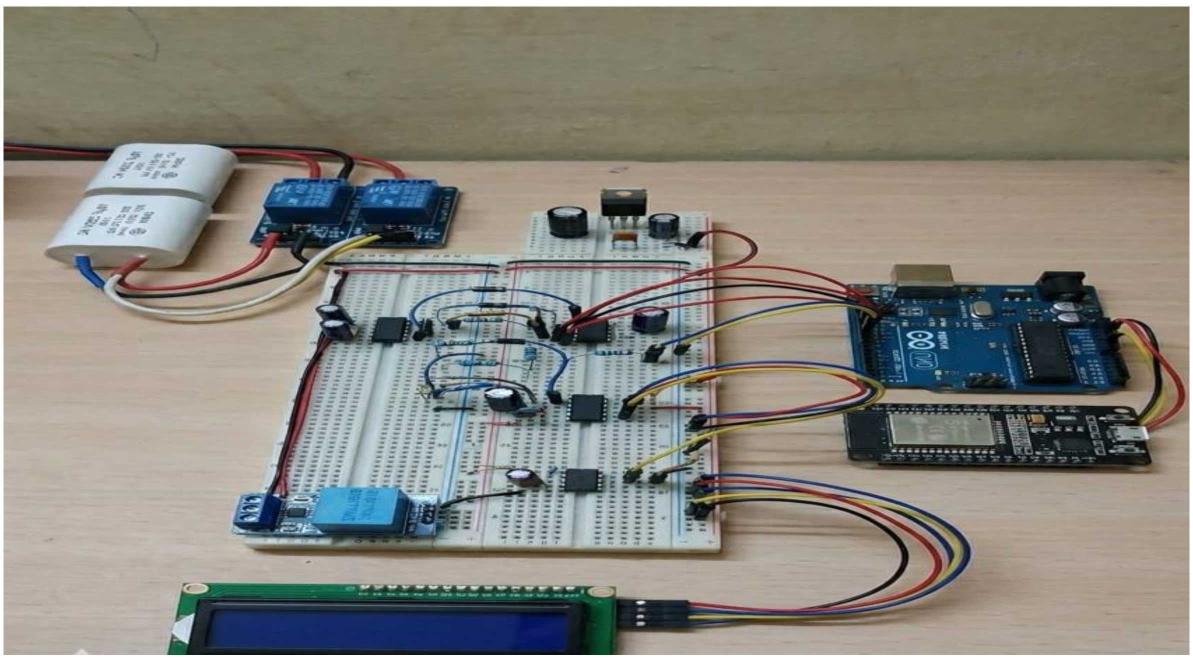
Hardware Circuit Implemented:

The hardware prototype of the Power Factor Estimation and Correction with Real-Time Energy Tracking system is composed of a number of functional modules working in tandem to provide an efficient power factor measurement and correction. The power supply of the system is achieved through a regulated power supply unit, which includes a 230V/12V transformer for reducing AC voltage. The AC voltage is then converted to DC through an IN4007 bridge rectifier and filtered through a 100µF capacitor to eliminate ripples. Regulated voltages are achieved through IC 7812 and IC 7805, which provide +12V and +5V voltages required for the proper functioning of sensors, microcontrollers, and other components.

The sensor section of the system includes a ZMPT101B sensor connected in parallel with the AC supply through a resistor network for safe voltage measurement. Similarly, an ACS712-5A sensor is connected in series with the load to measure current flow. These sensor outputs are then sent to signal conditioning circuits through IC 741 op-amp circuits working in non-inverting mode. In these circuits, a 10KΩ and 100KΩ resistor network is used to achieve desired gain.

The processing unit contains an Arduino UNO, which is used for analog-to-digital conversion as well as for calculating parameters such as RMS value and phase difference, whereas the ESP32 is used for executing the PFC logic as well as for enabling the transmission of data through Wi-Fi. The correction unit contains three 5V relay modules for switching three 10μF/250V AC capacitors, which are connected in parallel with the load for improving the efficiency of the system, depending on the PFC condition.

Also, a 16×2 LCD display with an I2C module is used for displaying real-time electrical parameters such as voltage, current, and power factor. The LCD display is connected to the Arduino UNO using SDA and SCL pins, which reduces the complexity of the system by minimizing the number of wires used, thus making it compact, user-friendly, and suitable for use in various applications.



Hardware Results:

Measured parameters on hardware prototype (resistive-inductive load, 60W motor):

Parameter	Before Correction	After Correction	Change
Power Factor (PF)	0.851 (lagging)	0.978 (near unity)	+14.9%
Line Current (IRMS)	1.39 A	1.18 A	−15.1%
Reactive Power (Q)	178.4 VAR	58.2 VAR	−67.4%
Active Power (P)	318.6 W	318.6 W	No change
Apparent Power (S)	374.2 VA	325.8 VA	−12.9%

Inference:

It is clear that the results achieved from the implementation of the Power Factor Estimation and Correction with Real-Time Energy Tracking system show its effectiveness in improving the power quality and efficiency of the system. The system has improved the power factor from 0.85 to 0.98, which is in line with the industrial requirement of maintaining the power factor greater than or equal to 0.95. This shows that the system is effective in reducing the phase difference between the voltage and current, which is essential in ensuring the efficient use of electrical energy.

Besides that, the system is also effective in reducing the reactive power by more than 67%, which is essential in reducing the apparent current in the system. The reduced current minimizes the I^2R losses in the transmission and distribution lines, which is essential in ensuring the efficient use of electrical energy and reducing the energy wastage in the system. The implementation of the three-stage capacitor bank in the system is also essential in ensuring the efficient use of electrical energy in the system. The system is also more advanced and user-friendly due to the integration of the real-time LCD and ESP32-based Wi-Fi communication in the system.

Applications:

1. Industrial motor drive systems requiring automatic reactive power compensation to minimize electricity tariff penalties.

2. Smart home energy management systems to monitor and minimize electricity bills by optimizing the power factor of domestic appliances.

3. Distribution substations and feeders requiring real-time monitoring and automatic correction of reactive power.

4. Educational laboratory demonstration of power quality, reactive power, and power factor correction principles.

5. IOT-based remote energy auditing systems for commercial buildings and factories.